

Orbital CDN: A Physically-Grounded, Predictive Control Plane for Content Delivery over LEO Constellations

Udit Jain

`orbital-cdn.vercel.app` · `github.com/uditjainstjis/orbital-cdn`

Abstract—Low-Earth-orbit (LEO) constellations are frequently proposed as a substrate for content delivery on the grounds that the vacuum speed of light exceeds the group velocity in silica by roughly 47%. This argument is necessary but not sufficient: the ascent and descent legs impose a fixed geometric overhead, so an orbital path is advantageous only beyond a crossover separation. We present a routing and control plane that (i) computes end-to-end latency from the actual geometry of the selected path rather than from a distance surrogate, (ii) evaluates rain-induced link degradation using a faithful implementation of the ITU-R P.838-3/P.839-4/P.618-13 recommendation chain, validated against all 28 published reference coefficients, (iii) expresses every competing routing objective in a single *latency-equivalent* currency so that policy weights trade commensurate quantities, and (iv) closes an observe–predict–plan–act–verify loop in which a gradient-boosted forecaster, trained on four months of NASA POWER reanalysis precipitation for eight real teleport sites, prices the probability of imminent link outage into the routing cost. Under a paired four-arm evaluation over 5,600 requests and 2,800 hours of recorded weather, request failures fall from 80 (weather-blind) to 2 (instantaneous observation) to 0 (forecast-aware). We report two negative results that qualify the contribution: the dominant share of the benefit accrues from *observation* rather than *prediction*, and our hysteretic agent attains no failure reduction over stateless predictive routing while incurring +5.3 ms of mean latency. Independently, the simulator reproduces a published LEO-versus-fibre crossover distance of 4,472 km without having been calibrated against it.

Keywords—LEO constellations, content delivery networks, multi-objective routing, ITU-R P.618, rain attenuation, probabilistic forecasting, calibration, autonomous network operations.

1 Introduction

The case for routing content through low-Earth orbit rests on a single physical asymmetry. Light propagates through vacuum at $c_0 = 299,792 \text{ km/s}$ but through fused-silica fibre at $c_f = c_0/n \approx 204,190 \text{ km/s}$ for a refractive index $n \approx 1.468$, a speed advantage of approximately 47%. A path that ascends to a satellite, traverses a sequence of optical inter-satellite links (ISLs), and descends to a ground station may therefore deliver a lower round-trip time (RTT) than a terrestrial path between the same endpoints.

This argument is incomplete. The ascent and descent legs contribute a separation-independent overhead of order $2h/\sin\theta$ for shell altitude h and elevation angle θ . Consequently the orbital path is advantageous only when the terrestrial separation is large enough to amortise that overhead. The engineering question is therefore not *whether* orbit is faster, but *where the crossover lies* and *what else is sacrificed* in reaching it—because a LEO content plane is subject to three constraints with no terrestrial analogue: orbital eclipse, which removes solar power from a compute-bearing satellite; transit of the South Atlantic Anomaly (SAA), where the inner Van Allen belt descends to approximately 200 km; and rain fade at the ground segment, which above 10 GHz can exceed any economically closable link margin.

1.1 Contributions

1. **A standards-faithful attenuation model.** We implement the ITU-R recommendation chain exactly and validate the implementation against all 28 published reference coefficients spanning 1–30 GHz (Section 4). Rain fade given a rain rate is an analytic problem; we treat it as such rather than approximating it with a learned surrogate.
2. **A commensurable cost functional.** All routing terms—propagation, eclipse, radiation, rain fade, learned penalties and predicted outage—are expressed in latency-equivalent milliseconds, eliminating a class of unit pathology we encountered and document in Section 6.
3. **A calibration-first predictive layer.** We show that a persistence baseline is a strong *discriminator* of imminent fade but a poor *probability estimator*, and that the learned model’s contribution is almost entirely calibration—the property a cost-weighted router requires (Section 8).
4. **A paired evaluation that decomposes the benefit.** By separating an arm that observes current conditions from one that forecasts, we attribute 97.5% of the achievable failure reduction to observation and the remainder to prediction (Section 11).
5. **Reported negative results.** Our hysteretic agent does not improve on stateless predictive

routing in this regime, and we state the conditions under which it would (Section 12).

2 Related Work

Handley [1] established that a LEO relay network can undercut terrestrial fibre on long-haul paths, reporting an advantage beyond roughly 3,000 km for a 1,150 km shell. Chaudhry and Yanikomeroglu [2] formalise the crossover distance as a function of shell altitude and fibre refractive index, obtaining 4,472 km at $h = 550$ km, $n = 1.4675$; critically, their terrestrial model assumes great-circle routing without path inflation, so realised crossovers should be shorter. We use their figure as an external check rather than a fitted parameter.

Rain attenuation on Earth–space paths is standardised in ITU-R P.618 [3], with specific attenuation coefficients in P.838 [4] and rain height in P.839 [5]. These are the models used in operational link design; we adopt them unmodified.

Orbital propagation follows the SGP4 analytic theory [6] applied to two-line element sets. Pfandzelter and Bermbach [7] characterise the reliability envelope of in-orbit edge computation, including single-event-upset rates under SAA transit, and provide the shell parameters we scale from.

3 System Model and Notation

Let $\mathcal{S}$ denote the satellite set, $|\mathcal{S}| = 180$, arranged as a Walker-Delta constellation of 9 planes $\times$ 20 satellites at inclination 53° and altitude $h_s = 550$ km. This is a 1/8-scale reduction of the authorised Starlink Gen1 shell 1 (1,584 satellites, 72×22 , identical altitude and inclination), adopted for real-time rendering tractability. A subset $\mathcal{D} \subset \mathcal{S}$, $|\mathcal{D}| = 4$, carries compute at $h_d = 640$ km. The ground segment comprises gateways $\mathcal{G}$, $|\mathcal{G}| = 8$, located at published teleport coordinates, and origin cities $\mathcal{C}$, $|\mathcal{C}| = 8$.

Satellite states are obtained by SGP4 propagation of two-line elements retrieved from CelesTrak, converted from the ECI frame to geodetic coordinates through Greenwich Mean Sidereal Time. When the catalogue is unavailable the system falls back to a closed-form Walker-Delta model,

$$\phi(t) = \arcsin(\sin i \sin \nu(t)), \quad (1)$$

$$\lambda(t) = \Omega(t) + \text{atan2}(\cos i \sin \nu(t), \cos \nu(t)), \quad (2)$$

with inclination i , true anomaly ν advancing at the orbital rate, and right ascension of the ascending node Ω advancing at the Earth-rotation rate.

Table 1 collects the principal symbols.

Table 1: Principal notation.

Symbol	Meaning
$g(\cdot, \cdot)$	great-circle distance (km)
$\tau(\cdot, \cdot)$	round-trip propagation time (ms)
θ	path elevation angle (deg)
γ_R	specific rain attenuation (dB/km)
A_p	attenuation exceeded for $p\%$ of a year (dB)
M	rain-fade link margin (dB)
Π	instantaneous outage probability
$w = (w_\ell, w_s, w_r, w_x)$	policy weights
ρ	learned penalty in $[0, 1]$
κ	forecast confidence in $[0, 1]$
Φ	failure cost, latency-equivalent (ms)

4 Link Physics

4.1 Specific attenuation (ITU-R P.838-3)

The specific attenuation obeys the power law

$$\gamma_R = k R^\alpha \quad [\text{dB/km}], \quad (3)$$

for rain rate R in mm/h. The polarisation-resolved coefficients are fitted as sums of Gaussians in $\log_{10} f$ with f in GHz. For k ,

$$\log_{10} k_\pi = \sum_{j=1}^4 a_j \exp \left[- \left(\frac{\log_{10} f - b_j}{c_j} \right)^2 \right] + m_k \log_{10} f + c_k, \quad (4)$$

and for α , with five terms and *without* the outer exponentiation,

$$\alpha_\pi = \sum_{j=1}^5 a_j \exp \left[- \left(\frac{\log_{10} f - b_j}{c_j} \right)^2 \right] + m_\alpha \log_{10} f + c_\alpha, \quad (5)$$

where $\pi \in \{H, V\}$ indexes polarisation. The asymmetry between (4) and (5)— k is fitted in logarithmic space and must be exponentiated, α is not—is a common source of silent implementation error, since a transposed convention still yields plausible-magnitude output.

For arbitrary path geometry and polarisation tilt τ (45° for circular polarisation),

$$k = \frac{1}{2} [k_H + k_V + (k_H - k_V) \cos^2 \theta \cos 2\tau], \quad (6)$$

$$\alpha = \frac{k_H \alpha_H + k_V \alpha_V + (k_H \alpha_H - k_V \alpha_V) \cos^2 \theta \cos 2\tau}{2k}. \quad (7)$$

Validation. Equations (4)–(7) are evaluated against the reference table published in the recommendation itself. All 28 coefficients across $\{1, 12, 19, 20, 25, 28, 30\}$ GHz reproduce to within 2% (Table 2); the check executes as a unit test, so a transcription error cannot pass unnoticed.

The four-order-of-magnitude spread in k between 1 GHz and 20 GHz is not merely a validation artefact: it

Table 2: Validation of the P.838-3 implementation against the published reference table. All 28 checks pass within 2%.

f (GHz)	k_H	α_H	k_V	α_V
1	0.0000259	0.9691	0.0000308	0.8592
12	0.02386	1.1825	0.02455	1.1216
19	0.08084	1.0691	0.08642	0.9930
20	0.09164	1.0568	0.09611	0.9847
25	0.1571	0.9991	0.1533	0.9491
28	0.2051	0.9679	0.1964	0.9277
30	0.2403	0.9485	0.2291	0.9129

is the quantitative reason that hydrometeor attenuation is negligible at UHF and dominant at Ka-band, and it disqualifies UHF-band packet corpora as training data for Ka-band fade prediction. We return to this in Section 13.

4.2 Rain height and slant geometry

Rain height follows P.839-4 as $h_R = h_0 + 0.36$ km, where h_0 is the mean annual 0°C isotherm height. We use the latitude parameterisation $h_0 = 5.0 - 0.075(|\phi| - 23)$ km for $|\phi| \geq 23^\circ$ and 5.0 km otherwise, in place of the gridded map, and flag this as an approximation.

The P.618-13 slant-path procedure then proceeds as follows. For $\theta \geq 5^\circ$ the path length below the rain height is

$$L_s = \frac{h_R - h_g}{\sin \theta}, \quad L_G = L_s \cos \theta, \quad (8)$$

with h_g the station elevation. The horizontal reduction factor accounts for the finite spatial extent of a rain cell,

$$r_{0.01} = \left[1 + 0.78 \sqrt{\frac{L_G \gamma_R}{f}} - 0.38 (1 - e^{-2L_G}) \right]^{-1}, \quad (9)$$

and the vertical adjustment factor, with $\zeta = \arctan[(h_R - h_g)/(L_G r_{0.01})]$ and $\chi = \max(0, 36 - |\phi|)$,

$$\nu_{0.01} = \left[1 + \sqrt{\sin \theta} \times \left(31(1 - e^{-\theta/(1+\chi)}) \frac{\sqrt{L_R \gamma_R}}{f^2} - 0.45 \right) \right]^{-1}. \quad (10)$$

The effective path length is $L_E = L_R \nu_{0.01}$ and the attenuation exceeded for 0.01% of an average year is

$$A_{0.01} = \gamma_R L_E. \quad (11)$$

Extrapolation to other exceedance percentages $p \in [0.001, 5]\%$ uses

$$A_p = A_{0.01} \left(\frac{p}{0.01} \right)^{-\beta}, \quad (12)$$

$$\beta = 0.655 + 0.033 \ln p - 0.045 \ln A_{0.01} - \varsigma(1 - p) \sin \theta, \quad (13)$$

with ς vanishing for $p \geq 1\%$ or $|\phi| \geq 36^\circ$.

4.3 Instantaneous fade and outage

For an observed rain rate the same specific attenuation and horizontal reduction are applied to the current rate rather than the annual statistic, yielding an instantaneous fade $A(t)$. We note explicitly that (3) is valid pointwise for any R , whereas the P.618 *statistics* presuppose a one minute integration time; our observations are hourly reanalysis means, which attenuate peak rates. The resulting fade estimate is therefore *conservative*, biased against the orbital system rather than for it.

Rather than thresholding fade against margin, we map the residual margin through a logistic link,

$$\Pi(t) = \sigma \left(\frac{A(t) - M}{s} \right) = \left[1 + \exp \left(\frac{M - A(t)}{s} \right) \right]^{-1}, \quad (14)$$

with $M = 6$ dB and softness $s = 1$ dB. The smooth transition encodes estimator uncertainty: a hard threshold would assert a precision the hourly-mean input does not support.

5 Path Geometry and Latency

Great-circle separation uses the haversine form

$$g(a, b) = 2R_\oplus \arcsin \sqrt{\sin^2 \frac{\Delta\phi}{2} + \cos \phi_a \cos \phi_b \sin^2 \frac{\Delta\lambda}{2}}. \quad (15)$$

Ground-to-satellite legs use the slant range $\sqrt{g^2 + h^2}$; inter-satellite legs are measured as arcs at orbital radius, $g(R_\oplus + h)/R_\oplus$.

The complete request path is

$$\underbrace{c \rightarrow u}_{\text{slant}} \rightarrow \underbrace{u \rightarrow \dots \rightarrow d}_{\text{ISL arcs}} \rightarrow \underbrace{d \rightarrow u'}_{\text{ISL arc}} \rightarrow \underbrace{u' \rightarrow \eta}_{\text{slant}}, \quad (16)$$

for origin c , uplink satellite u , data centre $d \in \mathcal{D}$, a relay u' in view of gateway $\eta \in \mathcal{G}$, terminating at the gateway. Two properties of this construction deserve emphasis. First, the descent is routed via a satellite *actually in view of the gateway*; a direct $d \rightarrow \eta$ slant range would traverse the planetary interior whenever d lies on the far side. Second, the chain already egresses and returns to the ground, so its length is the complete request-response path and must *not* be doubled to obtain RTT. Both are errors we committed and corrected; the second inflated one origin's path by 45,000 km.

End-to-end latency is then

$$\text{RTT} = \frac{\ell_{\text{path}}}{c_0} \cdot 10^3 + T_{\text{proc}} + T_{\text{wx}}, \quad (17)$$

with compute term T_{proc} and rain-fade delay T_{wx} .

5.1 Terrestrial baseline

The comparison baseline is *measured*, not modelled: ICMP round-trip times between third-party vantage points, cross-checked against a public cloud provider's published 30-day median backbone latencies (Table 3).

Table 3: Measured terrestrial baselines against the physical model. Distances are to the nearest terrestrial cloud region.

Origin	g (km)	Measured	Backbone	Model
New York	325	7.5	8	15
London	637	14.4	17	21
Delhi	4,145	92.1	53	79
Dubai	4,836	122.2	100	91
Lagos	4,873	118.5	—	92
Tokyo	5,322	66.4	72	99
Sydney	6,305	92.7	95	116
São Paulo	7,626	113.1	118	138

These are datacentre-to-datacentre and therefore exclude consumer access latency, making the baseline conservative. A physical model,

$$\text{RTT}_f = 2 \left(\frac{\varrho g}{c_f} \cdot 10^3 + \varsigma_{sw} \frac{\varrho g}{10^3} + T_{acc} \right) + T_{\text{proc}}, \quad (18)$$

with route-inflation factor $\varrho = 1.42$, switching cost $\varsigma_{sw} = 1$ ms per 1,000 km and access latency $T_{acc} = 5$ ms, is retained as a fallback for origins outside the measurement set and is displayed alongside the measurement so that model error is visible rather than concealed.

6 Multi-Objective Routing in a Commensurable Currency

Data-centre and gateway selection minimise scalarised costs in which *every* term carries units of latency-equivalent milliseconds:

$$\begin{aligned} C_{\mathcal{D}}(c, d) &= w_\ell \tau(c, d) + w_s \kappa_e \mathbf{1}[\text{ecl}(d)] \\ &\quad + w_r \kappa_a \mathbf{1}[\text{saa}(d)] + w_\ell \kappa_p \varsigma \rho_d^\ell \\ &\quad + w_r \kappa_p \varsigma \rho_d^r, \end{aligned} \quad (19)$$

$$\begin{aligned} C_{\mathcal{G}}(c, \eta) &= w_\ell \tau(c, \eta) + w_x T_{\text{wx}}(\eta) \\ &\quad + w_x \kappa_p \varsigma \rho_\eta^x + w_x J(\eta), \end{aligned} \quad (20)$$

with $\tau(a, b) = 2g(a, b)/c_0 \cdot 10^3$, eclipse cost $\kappa_e = 25$ ms, radiation cost $\kappa_a = 40$ ms, adaptation scale $\kappa_p = 30$ ms, adaptation gain $\varsigma = 0.6$, and predictive term J defined in (24). Selection is arg min over each candidate set.

Why commensurability is load-bearing. An earlier formulation mixed a distance normalised to $[0, 1]$ with indicator penalties of unit magnitude. Under a balanced weighting the eclipse indicator then dominated half a planetary circumference of detour, and the router dispatched requests to the antipodal hemisphere in pursuit of insolation. Stating the eclipse penalty as “25 ms of detour at unit weight” renders the trade explicit and auditable; it also makes the weights interpretable to an operator, which a normalised formulation does not.

Emergent policy structure. With four weight vectors the resulting operating points form a Pareto frontier that was not designed (Table 4). Notably, the reliability-weighted policy attains *exactly* zero rain exposure—an unspecified consequence of $w_x = 0.90$ —at a cost of 12 ms in median latency.

Table 4: Emergent operating points. Rain exposure is the fraction of served requests whose gateway was in precipitation.

Policy	w_ℓ	$w_s/w_r/w_x$	p_{50} RTT	Rain exp.
latency	0.95	0.05/0.05/0.05	61 ms	13.7%
balanced	0.50	0.50/0.50/0.40	67 ms	13.9%
reliable	0.20	0.30/0.95/0.90	73 ms	0.0%
green	0.20	0.90/0.20/0.30	78 ms	5.5%

7 Adaptation from Observed Telemetry

Every served request appends a record to an append-only log. Over a user-selected window W , aggregates induce penalties $\rho \in [0, 1]$ that re-enter (19)–(20). For gateways,

$$\rho_\eta^x = \min \left(1, \frac{1}{|W_\eta|} \sum_{t \in W_\eta} T_{\text{wx}}(\eta, t) / T_{\text{wx}}^{\max} \right). \quad (21)$$

For data centres we separate two signals that carry different policy coefficients: an observed tail-latency excess relative to the best performer,

$$\rho_d^\ell = \min \left(1, \frac{q_{95}(d) - \min_{d'} q_{95}(d')}{\min_{d'} q_{95}(d')} \right), \quad (22)$$

and an observed SAA-transit fraction ρ_d^r .

An information-theoretic exclusion. We deliberately do *not* learn a historical eclipse fraction. The cost function already reads $\mathbf{1}[\text{ecl}(d)]$ instantaneously; a windowed average of the same deterministic quantity is strictly less informative than the indicator it would accompany, and charging both double-counts identical orbital geometry. The general principle we adopt is that a learned term is admissible only if it encodes information not available to the controller by direct observation. Under this criterion gateway weather qualifies—it is genuinely stochastic—whereas eclipse, being deterministic celestial mechanics, does not.

Effect. Toggling adaptation alters the selected route in 6 of 32 origin–policy combinations; varying the analysis window alters it in 4/32 (7d versus 30d) and 5/32 (24h versus 7d).

8 Predictive Layer

8.1 Problem statement

Let $A_\eta(t)$ denote fade at gateway η . We estimate

$$p_h(\eta, t) = \Pr \left[\max_{t < t' \leq t+h} A_\eta(t') > A^* \mid \mathcal{F}_t \right], \quad (23)$$

for horizons $h \in \{1, 3, 6, 12\}$ hours and threshold $A^* = 3$ dB, where $\mathcal{F}_t$ is the filtration generated by observations up to t .

Table 5: Held-out forecast performance. Persistence dominates on ranking at long horizons; the learned model dominates on calibration at every horizon.

Horizon	Persistence		Gradient boosting	
	AUC	Brier	AUC	Brier
+1h	0.998	0.0294	0.999	0.0030
+3h	0.990	0.0290	0.991	0.0071
+6h	0.973	0.0307	0.949	0.0139
+12h	0.944	0.0329	0.892	0.0271

8.2 Data and leakage control

Precipitation is NASA POWER hourly reanalysis for the eight teleport coordinates, 2,928 hours per site. We note a unit convention that is easy to mishandle: the hourly endpoint reports PRECOTTCORR in mm/day despite hourly sampling, requiring division by 24; omitting this inflates every rate by 24 $\times$ and renders all downstream fade estimates meaningless.

Twenty-one features are constructed, all measurable at or before t : lagged rates at $\{1, 2, 3, 6, 12\}$ hours, first differences over 1 and 3 hours, rolling means over $\{3, 6, 24\}$ hours, rolling maxima over $\{6, 24\}$ hours, a 24-hour wet fraction, instantaneous fade, hours since last wet, site climatology, absolute latitude, and diurnal and seasonal phase.

Partitioning is *temporal*, never random: consecutive hours are strongly dependent, and a random split leaks future information into training. Training uses hours $[0, 1756)$, validation $[1756, 2196)$, and test $[2196, 2928)$. A second partition holds out entire *sites* to assess transfer to a gateway never observed.

8.3 Results and the primacy of calibration

Table 5 reports held-out performance against three baselines. The learned model is a gradient-boosted ensemble (90 estimators, 12 leaves, depth 5), exported to 71 KB of JSON and evaluated client-side.

The pattern is unambiguous and, we argue, is the substantive finding of this section. Persistence is an excellent *ordering* of imminent fade—it is difficult to beat on AUC, and at 6 and 12 hours we do not beat it—but it is a poor *probability*. Brier score improves by factors of 2 to 10 at every horizon. Because the controller multiplies p_h by a cost in (24) rather than merely sorting by it, calibration and not discrimination is the operative property. A system that ranks perfectly but reports 0.9 where the truth is 0.3 will systematically over-purchase latency to avoid failures that were never likely.

Held-out AUC on entirely unseen sites is 0.999, 0.994, 0.987 and 0.958 across the four horizons, indicating that the learned structure is not merely site memorisation.

8.4 Predictive cost and confidence gating

The predicted-failure term entering (20) is

$$J(\eta) = \Phi [\Pi_\eta + \kappa(\eta) \bar{p}(\eta) (1 - \Pi_\eta)], \quad (24)$$

where $\bar{p} = \sum_h \omega_h p_h / \sum_h \omega_h$, $\Phi = 900$ ms, and horizon weights $\omega = (1.0, 0.55, 0.25, 0.10)$. Only the forecast is attenuated by confidence κ ; the present observation Π_η is certain and enters undamped.

Confidence is reduced when the operating point departs from the training support—an out-of-distribution rate of 90 mm/h at a site whose recorded maximum is 11 mm/h yields $\kappa = 0.45$, below the autonomy threshold $\kappa_{\min} = 0.60$. Below that threshold the agent may recommend but not act.

Degradation policy. If the predictive subsystem raises, times out, or is disabled, $J \equiv 0$ and the deterministic controller of Section 6 is recovered unchanged. Learned components are never a single point of failure for routing.

9 The Operations Agent

9.1 Formalisation

The agent is a deterministic policy engine—explicitly not a learned controller—executing on a fixed cadence decoupled from rendering. At tick t it observes the network state, evaluates candidate routes, and emits an action from $\{\text{HOLD}, \text{PROPOSE}, \text{REROUTE}\}$.

Define the horizon-weighted forward risk $\bar{r}(\eta) = \sum_h \omega_h p_h(\eta) / \sum_h \omega_h$, and let η_t be the incumbent gateway and $\eta^* = \arg \min_\eta C_G(c, \eta)$. The agent reroutes iff the conjunction

$$\Theta_t \wedge \Lambda_t \wedge \Xi_t \wedge \Delta C_t \geq \delta \wedge \Delta r_t \geq \epsilon \wedge \kappa \geq \kappa_{\min} \quad (25)$$

holds, where

$$\begin{aligned} \Theta_t &: \bar{r}(\eta_t) \geq r_{\text{trig}} \vee \Pi_{\eta_t} \geq \Pi_{\text{trig}} && (\text{trigger}), \\ \Lambda_t &: t - t_{\text{route}} \geq \Lambda_{\min} && (\text{dwell}), \\ \Xi_t &: t - t_{\text{act}} \geq \Xi_{\min} && (\text{cooldown}), \\ \Delta C_t &= C_G(c, \eta_t) - C_G(c, \eta^*) - \varpi && (\text{net gain}), \\ \Delta r_t &= \bar{r}(\eta_t) - \bar{r}(\eta^*) && (\text{risk drop}). \end{aligned}$$

Here ϖ is an explicit switching cost. Operationally $r_{\text{trig}} = 0.45$, $\Pi_{\text{trig}} = 0.30$, $\Lambda_{\min} = 20$ s, $\Xi_{\min} = 15$ s, $\delta = 25$ ms, $\epsilon = 0.15$, $\varpi = 12$ ms.

Stability. The dwell and cooldown predicates render the switching sequence non-Zeno by construction: consecutive actions are separated by at least $\min(\Lambda_{\min}, \Xi_{\min})$, bounding the switching rate independently of forecast volatility. The margin conditions δ, ϵ impose hysteresis, so a marginal cost inversion cannot induce oscillation between two gateways of near-equal cost—the failure mode that makes a naive greedy controller worse than no controller at all.

9.2 Authority and verification

Three authority levels are exposed: OFF (advisory suppressed), ASSIST (propose and await human confirmation), and AUTOPILOT (act). The default is ASSIST. This is a deliberate design position rather than an implementation limitation: no operational satellite programme presently delegates unsupervised manoeuvre or reconfiguration authority to a predictive model, and a system claiming otherwise would be dismissed by any reviewer with domain experience.

Every action is subsequently scored against realised conditions. For a decision taken at t with stated risk p , the realised outcome $y = \mathbf{1}[\max_{t < t' \leq t+h} A(t') > A^*]$ is recorded and the Brier score $\mathcal{B} = \frac{1}{N} \sum (p_i - y_i)^2$ and reliability diagram are maintained. A forecast that is never scored is an assertion, not a prediction.

10 Experimental Methodology

We evaluate under a *paired* design: all arms are exposed to an identical request schedule against an identical weather realisation, replayed from a common epoch. Without pairing, inter-arm variance would be dominated by which arm happened to execute during a storm.

Arms are distinguished solely by their observational endowment:

- A** BLIND — static weather label only; the pre-predictive controller.
- R** REACTIVE — observes instantaneous fade; no forecast.
- B** PREDICTIVE — observes fade and forecast.
- C** AUTOPILOT — as B, plus a per-origin standing route maintained by the agent of Section 9.

Isolating **R** from **B** is the methodological crux: it separates the value of *observing* from the value of *forecasting*, a decomposition that a three-arm design conflates and that determines whether a predictive subsystem is justified at all.

Outcome is adjudicated by physics identically in every arm: a request fails iff $A_\eta(t) > M$. The schedule comprises 5,600 requests over 2,800 recorded hours, with origins and policies drawn from a fixed pseudo-random sequence.

11 Results

Table 6 decomposes the benefit. Of 80 baseline failures, 78 (97.5%) are eliminated by instantaneous observation alone; forecasting eliminates the residual 2 at a cost of +0.2 ms in mean latency. The agent attains no further failure reduction and incurs +5.3 ms relative to arm B, executing 9 reroutes of which 7 (78%) preceded actual degradation of the abandoned gateway.

Table 6: Paired four-arm evaluation. 5,600 requests, 2,800 hours of recorded weather, identical schedule and realisation in every arm.

Arm	Failures	Rate	Mean (ms)	p_{95} (ms)
A BLIND	80	1.43%	4.0	12.7
R REACTIVE	2	0.04%	3.7	10.0
B PREDICTIVE	0	0.00%	3.9	11.6
C AUTOPILOT	0	0.00%	9.2	31.9

External validation. Using measured baselines, the system loses at London (637 km, 0%) and New York (325 km, 0%) and wins from Delhi (4,145 km, $\approx 90\%$) outward, placing its empirical crossover in the same region as the 4,472 km reported by [2] for an identical shell altitude. No parameter was fitted to that figure. Given that [2] models terrestrial paths without route inflation whereas our baseline is measured and therefore inflated, a realised crossover at or slightly below their value is the theoretically expected outcome.

Environmental bound. A caveat constrains the achievable effect size: across the four-month record only one of eight gateways experiences fade exceeding the 6 dB margin, in 19% of its hours (maximum 19.9 dB); three others do so in under 0.6% of hours and four never. The measurable benefit of any weather-aware routing is bounded above by this exposure.

12 Discussion

12.1 Observation dominates prediction

The decomposition in Table 6 does not support a strong claim for forecasting in this regime. Reporting only arms A and B would have licensed the statement “predictive routing eliminated 100% of failures”—literally true and materially misleading, since a substantially simpler reactive controller captures 97.5% of that benefit.

We attribute this to the temporal structure of the hazard. Rain cells at these sites persist over multiple hours, so the instantaneous state is already highly informative about the near future; this is the same property that makes persistence such a strong AUC baseline in Table 5. Forecasting should be expected to contribute materially only where the hazard has short autocorrelation relative to the actuation latency, or where actuation itself is slow enough that acting on the present is already too late.

12.2 When hysteresis is worth its cost

Arm C is dominated by arm B. The explanation is structural rather than parametric: in this simulator, re-selecting a gateway per request is costless, so a standing route with dwell and hysteresis constraints can only retain a suboptimal assignment between switching opportunities. The agent’s machinery prices a switching cost ϖ that the environment does not actually charge.

This identifies the condition under which the agent becomes justified rather than merely defensible: environments in which route changes carry genuine cost—session affinity, cache warmth, key re-establishment, signalling load—so that $\varpi > 0$ in reality and not only in the controller’s model. We regard the present result as correctly characterising the agent’s value as *zero in this regime*, and we report it rather than tuning the environment until the agent appears necessary.

12.3 On the admissibility of learned components

Two decisions in this work were resolved by the same criterion: a learned component is admissible only where it supplies information unavailable by direct observation or by an accepted analytic model. Rain fade given a rain rate is analytic and standardised, so it is computed, not learned. Historical eclipse fraction is deterministic and already observed, so it is excluded from the adaptation vector. Future precipitation is genuinely stochastic, so it is learned. We suggest this criterion generalises: systems that learn quantities they could compute or observe acquire the appearance of intelligence and the substance of noise.

13 Threats to Validity

Synthetic demand. Request traffic is simulated and labelled as such in the interface. Weather, physics and the comparison baselines are not. Generalisation of the routing decisions to a real demand distribution is untested.

Reanalysis smoothing. Hourly means attenuate the sub-hourly peaks that dominate Ka-band outage. Absolute fade magnitudes are therefore underestimated; the direction of the bias opposes our claims.

Scale reduction. The constellation is 1/8 of the shell it models. Inter-satellite link range limits—operationally of order 5,400 km—are not enforced, which at reduced density permits hops a real system could not close.

Geometric idealisation. The SAA is approximated by a rectangle over a region that is elliptical, bifurcating, and drifting westward at approximately $0.3^\circ/\text{yr}$. Rain height uses a latitude parameterisation rather than the gridded isotherm map.

Domain transfer. We evaluated and rejected a proposal to train the fade forecaster on a public corpus of UHF (≈ 437 MHz) satellite packet receptions. By Table 2, k differs by nearly four orders of magnitude between that band and Ka; the corpus cannot contain the hydrometeor signal the model would be required to learn. We record this as a methodological caution: dataset availability is not evidence of dataset relevance,

and the governing physics should be consulted before a corpus is adopted.

14 Reproducibility

The physics implementation is validated by an executable test reproducing all 28 published coefficients (`npm run test:itu`). Weather ingestion and model training are scripted end-to-end (`npm run data:weather`, `npm run train:fade`), with the trained ensemble exported as inspectable JSON. Held-out metrics, the temporal split boundaries, feature schema and data provenance are emitted as a machine-readable artefact alongside the model. The evaluation of Section 11 is deterministic under a fixed seed and executes in-browser.

15 Conclusion

We have described a control plane for LEO content delivery in which the physics is a faithful standards implementation rather than an approximation, the comparison baseline is externally measured rather than self-generated, and every routing objective is expressed in a single commensurable unit. The system reproduces an independently published crossover distance without having been fitted to it.

Its principal empirical finding is deflationary and, we believe, more useful for that: in a regime where the hazard is strongly autocorrelated, most of the attainable reliability benefit is obtained by *observing* current conditions, and a predictive subsystem contributes at the margin. The learned component’s value lies in calibration rather than discrimination, and the hysteretic agent’s value in this environment is nil. Establishing the conditions under which each becomes load-bearing—short hazard autocorrelation for the forecaster, non-zero switching cost for the agent—is the natural continuation of this work.

The broader claim we wish to make is not that orbital delivery is faster. It is that the boundary can be located, that locating it requires physics and measurement rather than assertion, and that a system which reports where it fails is more useful than one which reports only where it succeeds.

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